Micromac C

OPERATING MANUAL

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1.3 e					

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SYSTEA Srl

µMAC- C and µMAC- C MP
Instrument Maintenance Manual
Ver. 1.1 e

SUMMARY

1. INTRODUCTION

The **μMAC C** is an automatic chemical analyzer for on-line measurements, based on a new improvement of the exclusive analytical technology developed by SYSTEA named LFA* (**L**oop **F**low **A**nalysis);

μMAC C series, allows to run sequentially up to four chemistry on the same analyzer

Possible configurations available for μMAC C series are:

- **MP2**: two chemistries
- **MP3**: three chemistries
- **MP4**: four chemistries

The compactness and the robustness, together with the low reagents consumption, allows the system to be easy used directly for continuous monitoring.

(*) Patent pending

1.1 Analytical cycle

A μMAC C is able to run sequentially one or more analytical cycles (chemistries) under the control of the local microprocessor.

The necessary steps for a complete single analytical cycle are the following:

1. Sample aspiration and introduction in analytical reactor.
2. Sample blank measurement and colorimeter zeroing.
3. Reagents injection with the proper sequence and timing.
4. Sample and reagents mixing.
5. Heating if needed.
6. Optical density increase measurement with a double beam colorimeter.
7. End point measurement storing in non volatile RAM for successive remote and local readings.
8. Concentration value automatic calculation, based on the correlation with internal calibration factor.

The analytical cycle can be automatically repeated for another chemistries, up to four; under the microprocessor control that select a proper parameters set.

1.2 Calibration cycle

1. Calibration with working standard:

   The mixed working standard is normally placed inside the μMAC C; it will be used as it is, instead of sample, during calibration cycles. This is the standard procedure to calibrate the analyzer.

2. Calibration with stock calibrant and automatic dilution:

   An high concentration calibrant is placed inside μMAC C and is connected to one of valves between V1 and V7; to run a calibration cycle, the reactor is filled with distilled water, than a small quantity of the stock calibrant is injected in the reactor (standard addition) and diluted by mixing in the total reactor volume. This procedure is used only for special applications.

For each method, the calibration cycle analyze a standard to calculate the calibration factor; for MP configurations, a mixed calibrant is suggested.

SYSTEA

1.3 Technical data

Micromac 1000 & Micromac C series - Technical data

General description	Data
Sensor classification	Colorimetric dual beam with silicon detector or Ion selective electrode (only Micromac 1000 E)
Application	Seawater, surface water, potable water, waste water, industrial water, soils extract, beverages etc.

General specifications	Data
Power requirements	12 Vdc, power supply from 220Vac or other local supply to 12 Vdc available as option
Humidity	Up to 90% not condensable
Temperature range	10 – 40° C Analyzers < 25° Reagents
Unit dimensions and weight	Micromac 1000: 55 x 35 x 11 (d x h x w); 15 kg Micormac 1000 C/E/MP: 80 x 45 x 30 (h x w x d); 25 kg
Measuring points	Micromac 1000, C or E: 2 Micromac MP: 1 More measuring points available on request
Atmospheric pressure range	No limits
Effect of electromagnetic fields	EMC tested according CE compliance
Tolerance to electrostatic discharges	EMC tested according CE compliance
Pc Specification – O/S	PC 104 industrial standard Included
Positioning and installation details	Micromac 1000: bench top analyzer, to be installed at around 100 cm from floor, max 2 mt. from sampling point Micromac C/E/MP: wall mounting analyzer to be installed at 60/70 cm from floor, max 2 mt from sampling point
LFA reactor volume	10 ml

General specifications	Data (cont'd)
Materials in contact with sample	Glass. Silicone, plexiglass, stainless steel AISI 316
CE compliance	YES
Year 2000 compliance	YES
General hazards	Only chemical, for details see specific chemistries
Sample conditioning requirements	Depending on the matrix: Imhoff sedimentation stage or special filtering system between 10 and 50 micron in case of presence of not settleable sludge.

Sample delivery Operating ranges	Data
Temperature range	5°- 30 °C
Pressure	0.2 – 10 bar; Micromac 1000 max 0.5 bar
Flow	It depends from the sampling or filtering system; analyzer needs 20 ml of sample for cycle.
Connection	Micromac C/E/MP: Standard Teflon tube 1.6x3.2 mm other connection on request; Micromac 1000: nipple for silicon tube 1.8 x 4 mm.
Waste	Micromac C/E/MP: Pressure free 1.8 x 4 mm other on request; Micromac 1000: nipple for silicon tube 1.8 x 4 mm pressure free
Viscosity	5 cps
Turbidity	Not applicable; sample blank correction
Colour	Not applicable; sample blank correction
PH	Depending on the chemical method; normal 3 - 12

Signal inputs/outputs	**Data**
4 – 20 mA – Voltage 0 – 5	4 – 20 mA load 400 Ohm, linear response (galvanic isolator module available as option) or 0 – 5 V; separated for each stream and/or parameter in case of Micromac MP
Printer options	Optional, serial output RS232
Radio or modem links	Available as option
Earthing details	Not applicable, 12 Vdc power device
Alarms output	General alarm: 1 potential free switch (SPDT), max load 24V AC DC 0.5 A Limit signal: 1 potential free switch (SPDT), max load 24V AC DC 0.5 A Calibration alarm: 1 potential free switch (SPDT), max load 24V AC DC 0.5 A
Serial I/O for signals	Serial data output RS232 for bidirectional comunications, available as option
Analog Input	Analysis:1 digital contact with fotocoupler, galvanically isolated Calibration:1 digital contact with fotocoupler, galvanically isolated

Commissioning	**Data**
Manufactures set-up details / pre-installation guide	Available, supplied on order confirmation to allows preparing of installation site, also included in operating manual delivered with the analyzer.
Training program	Standard 2 days
Factory final test certification	Delivered with the analyzer
Delivery time	60 days
Description of sensor technology	Available, enclosed to the Instrument Maintenance Manual
Operating sequence	Available, enclosed to the LFA Software Manual
Calibration method	Available, enclosed to the LFA Software Manual

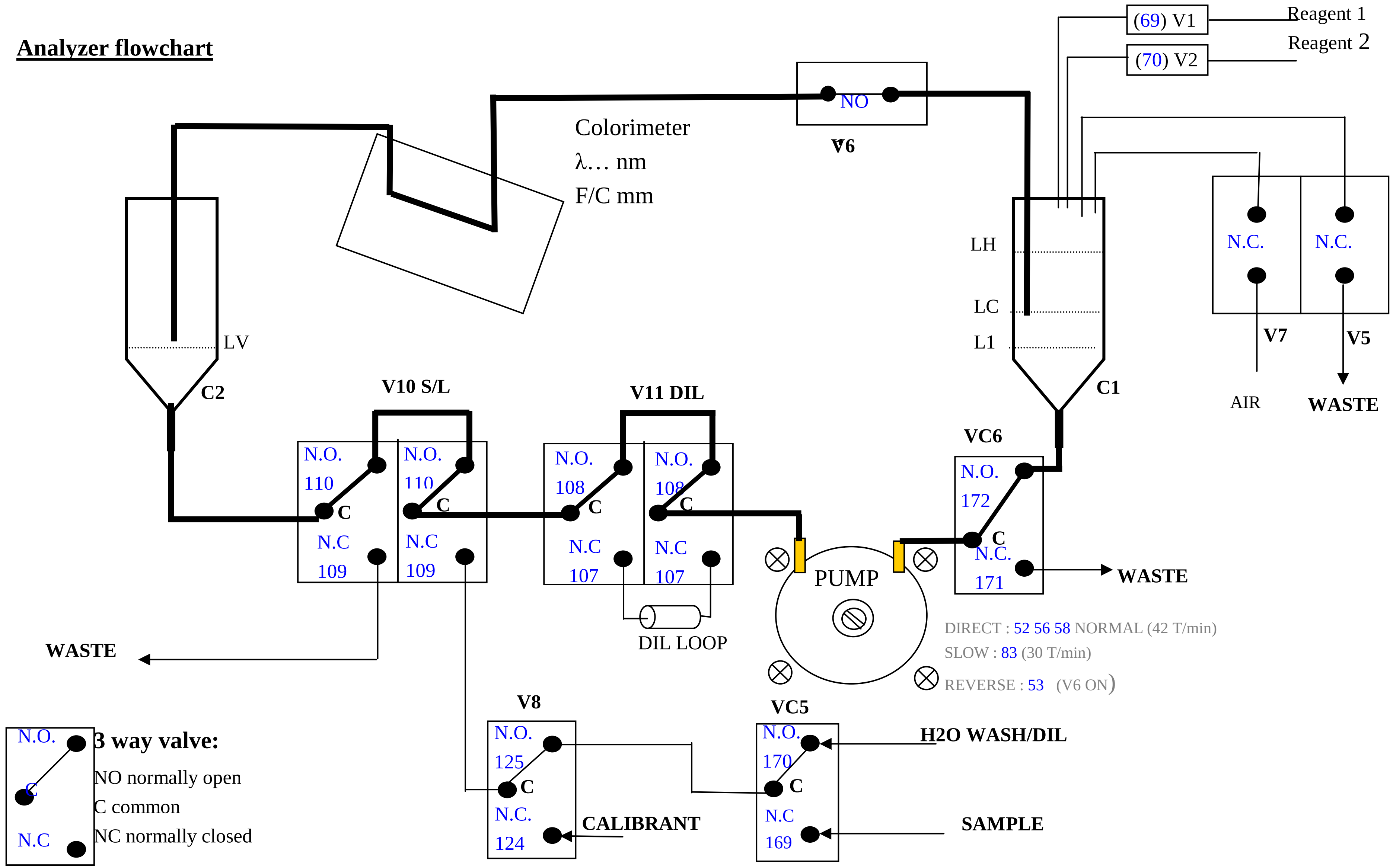
Operational specifications	Data
Type of analysis	Batch
Analysis interval	User selectable
Autodilution	Available: automatic for off scale samples or manually user selectable
Linearity	Over all specified range
Lowest detectable change	Depending on the method, normal 1% or 2% full scale
Repeatability	Depending on the method, normal better than 1% full scale
Response time	From 6 to 30 minutes depending on the method
Dead time	Not applicable, batch analyzer
Rise time	Not applicable, batch analyzer
Fall time	Not applicable, batch analyzer
Selectivity / interferences	As for official colorimetric method, or as for Ion selective electrode specifications
Drift in 24 hours	2% full scale
Memory effects	Not present, proper washing procedure

Operational calibration	Data
Frequency / intervals	Recommended: 24 hours for colorimetric equipment as for manufacturer specifications for electrodes.
Single / multi point	Multipoint: 0 and top of range
Matrix corrections	Yes, sample blank correction
Manual / automatic	Both
Handling / preparation	None
Sample introduction method	Integrated peristaltic pump

Quality checks	Data
Automatic/manual	Automatic
Frequency	During every calibration cycle, both automatic or manual
Percent of range	Standard top of range
System for control and actions	User selectable deviation warning, from the last calibration value

Maintenance	Data
Frequency schedule	2 weeks, reagent replacement if needed, visual inspection
Expected life of parts	Pump tube: 1500 hours of running Electrodes: 6 months or as specified from manufacturer Transmission tubing: 1 year
Reagent consumption	Colorimetric: max 200 microliters for test, usually 100/200 ml of single reagent is enough for one month of operation. Electrodes: buffers or ionic strength adapters as for sample matrix and manufacturer specifications.
Reagents stability	Depending on the method, from 15 to 30 days, temperature < than 25°C; for special applications reagent compartment can be refrigerated on request.
Manufacturer's service	Factory or on site service for repairs, special maintenance, upgrading or other; under warranty or not.
Operator maintenance task	Reagent replacement, pump tube replacement, cleaning procedures, visual inspections
List of consumables	Pump tube Tubing Kit
Indication of spares availability	2 weeks delivery time

1.4 Analyzer flowchart



2. DESCRIPTION

2.1 LFR reactor

A simple LFA monoparametric analytical reactor includes several devices (mechanical, hydraulics, optics) sequentially interconnected to form a ring defined LOOP FLOW REACTOR, in short LFR.

A standard LFR hydraulic schematic diagram is shown in the previous drawing; components shown are enough to run up to four chemistries, with the possibility to select two different wavelength. standard hydraulics also allows automatic dilution of off scale samples with a dilution ratio factory selected (usually 10)

Valves V10 S/L allows to open the LFR to pump sample, standard or blank solution in (SAMPLE position)or to keep the LFR closed in LOOP, to allows mixing reaction and so on. Valves Vdil are activated, when a sample dilution is needed.

The cylinder C1 allows the injection of the reagents and the calibrant (only in case of

calibration using a stock calibrant see 1.2 Calibration cycle) in the LFR, and contributes to create a strong turbulence inside the circuit, in order to speed up the mixing phase.

The valve V6, when activated, interrupts the LFR hydraulic connection and contributes to the vacuum production inside C1.

The pump P is a single tube peristaltic pump that can be activated in bi-directional direction

The valves VC5 and V8 make the selection of Sample, Diluent and Calibrant.

The cylinder C2 is normally filled with air and contributes to the vacuum production in C1, containing temporarily the volume of liquid pumped out from C1 by the pump P.

2.2 Sampling

With valves V10 S/L and VC5 NC, the pump P starts in direct mode; the sample is pumped trough V8, V10, V11, and VC6 in the cylinder C1, and once reached the level LC it goes up on the tube, passes through V6, through the colorimeters flow cells, and than reaches C2 and flows out to the waste through the opposite section of the valves V10 S/L.

After an appropriate sampling time, the cylinder C1 is filled up to the level LH by closing V6 and opening V5.

All the LFR components, including the flow cell, are now filled with sample; the system can read and store the Sample Blank value that will be subtracted from the final OD of the reaction product.

2.3 Analysis,

Following operation are valid for Sample analysis, Reagent Blank and Calibration

To start the analysis valves V10 S/L are switched NO to close LFR in LOOP.

2.3.1 Reagent injection

To inject reagents into the LFR the valve V6 is closed and the pump P is started in reverse mode until the liquid in C1 will reach the level Ll.

The volume of liquid between LH and Ll has been moved in C2 where has reached the level LV; the strong vacuum produced with this operation in C1 will allows to suck the reagents, with a thin and fast flow-out, activating for a precise time (proportional to the volume), the microvalves V1 or V2 connected to the reagents containers.

2.3.2 Pressure compensation

Opening the valve V6 will restore the original levels in C1 and C2.

To compensate the overpressure, after reagents injection, valves V10 S/L are switched NC for few seconds allowing discharge of excess of liquid contained in the circuit.

2.3.3 Mixing

Then valves V10 S/L are positioned again in LOOP (NO), the pump P is activated in direct mode, to allow a fast mixing of sample and the reagents.

The special position of the IN and OUT tubes inside C1 will contribute to produce a strong turbulence thus allowing a fast mixing and the start of the chemical reaction.

2.3.4 Reaction and OD reading

The reaction will take place in all the points of the reactor and therefore in the colorimeter flow cell , thus allowing the monitoring of the reaction from time 0 (reagents injection) to the end point.

If methods requiring heating, to speed up the colour development, tow types of heating bath can be included in the hydraulics:

1) Flow cell heater allowing to heat only the reaction product in the flow cell mainly used for ammonia
2) Manifold heater allowing to heat all the reaction product that flows inside the heating bath for a time defined in the analytical cycle.

At the reaction end point the analyzer measures and store the reaction O.D.

2.3.5 Calculations

Reagent Blank OD : stored during calibration and used to calculate the calibration factor

Calibrant OD: stored during calibration and used to calculate the calibration factor

OD Start: Sample blank OD

OD End: Sample OD (Final OD – Sample blank OD)

Sample concentration = OD end x Calibration factor

2.3.6 Wash

Valves V10 S/L are switched NC, all other valves are NO , to allows pumping of diluent (usually distilled water). Pump P starts in direct mode, water fill up LFR washing hydraulics.

After a proper washing time (about 1-2 min) LFR, remains in stand by filled with distilled water ready for a new run.

2.4 Calibration cycle

Allows the instrument recalibration using a working calibrant corresponding to the higher expected sample value or using a stock calibrant that will be automatically diluted during calibration cycle (see 1.2 calibration cycle)

2.4.1.1 Automatic calibration using working calibrant

Calibrant is pumped inside the LFR as follows:

with valves V10 S/L and V8 NC, the pump P starts in direct mode; calibrant is pumped trough V8, V10, V11 and VC6, in the cylinder C1, and once reached the level LC it goes up on the tube, passes through V6, through the colorimeters flow cells, and than reaches C2 and flows out to the waste through the opposite section of the valves V10 S/L.

After an appropriate sampling time, the cylinder C1 is filled up to the level LH by closing V6 and opening V5.

All the LFR components, including the flow cell, are now filled with sample; the system can read and store the Calibrant Blank value that will be subtracted from the final OD of the reaction product.

Calibrant analysis for calibration cycle follow same operations described in 2.3 Analysis

2.4.1.2 Recalibration using stock calibrants

The analyzer dilute automatically concentrated calibrant contained in the reagents compartment.

To produce the working calibrant it is used following procedure:

LFR is filled with diluent, valves S/L is switched off and the stock standard is injected using the same procedure used for reagent injection.

The microvalve connected to the calibrant container is activated for a predefined time and, therefore, a precise and reproducible calibrant volume is injected in LFR.

A mixing step starts allowing a fast diffusion of the injected volume in the total LFR volume.

With this technique is possible to obtain a working calibrant up to 200 time less concentrated than the stock calibrant.

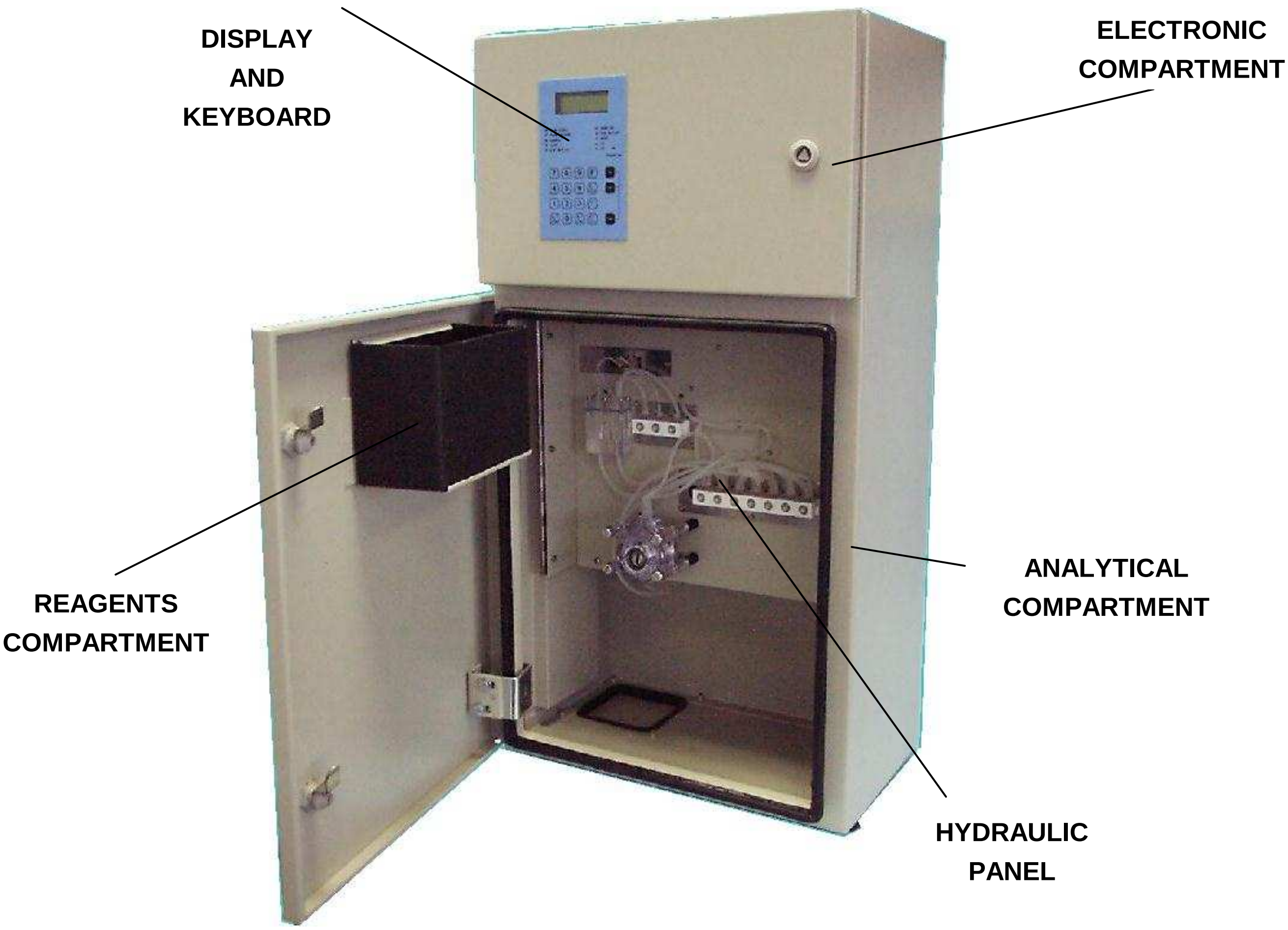
The working calibrant just obtained is analysed using the previously described analytical cycle and the O.D. values measured will be used for the calculation of the calibration coefficient.

3. INSTALLATION

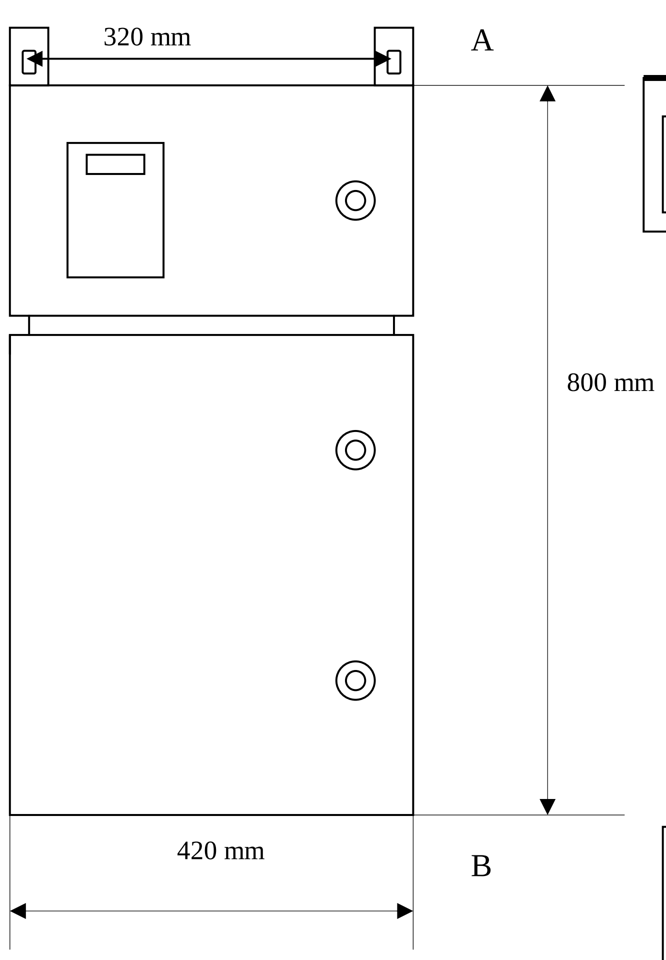
3.1 Physical dimensions and space requirement

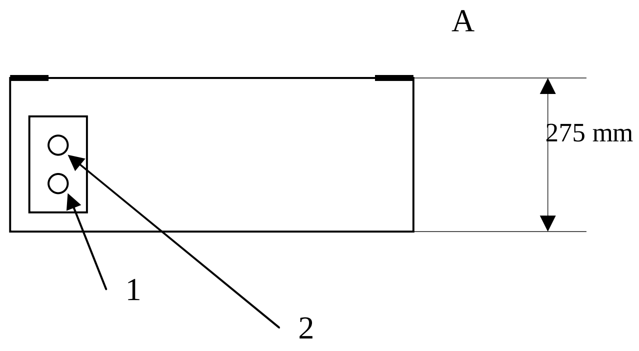
A picture of a µMAC-1000 C MP unit with main subsystems description is shown below; the physical dimensions are the following: 800 (H) x 420 (W) x 275 (D) mm.

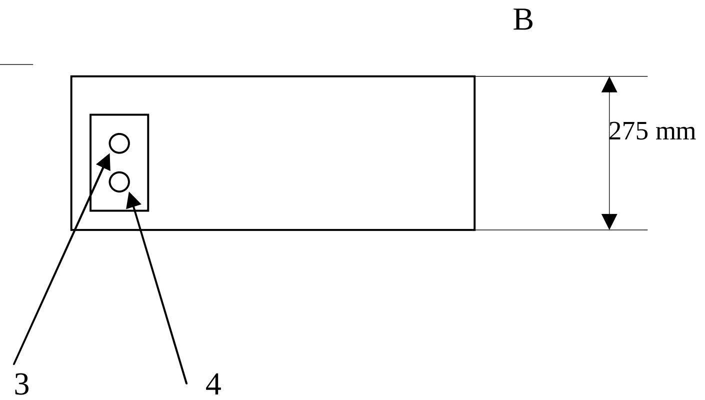
Including the reagents and the optional internal battery the µMAC-1000 C MP weights about 25 Kg.



3.2 Dimensional drawing





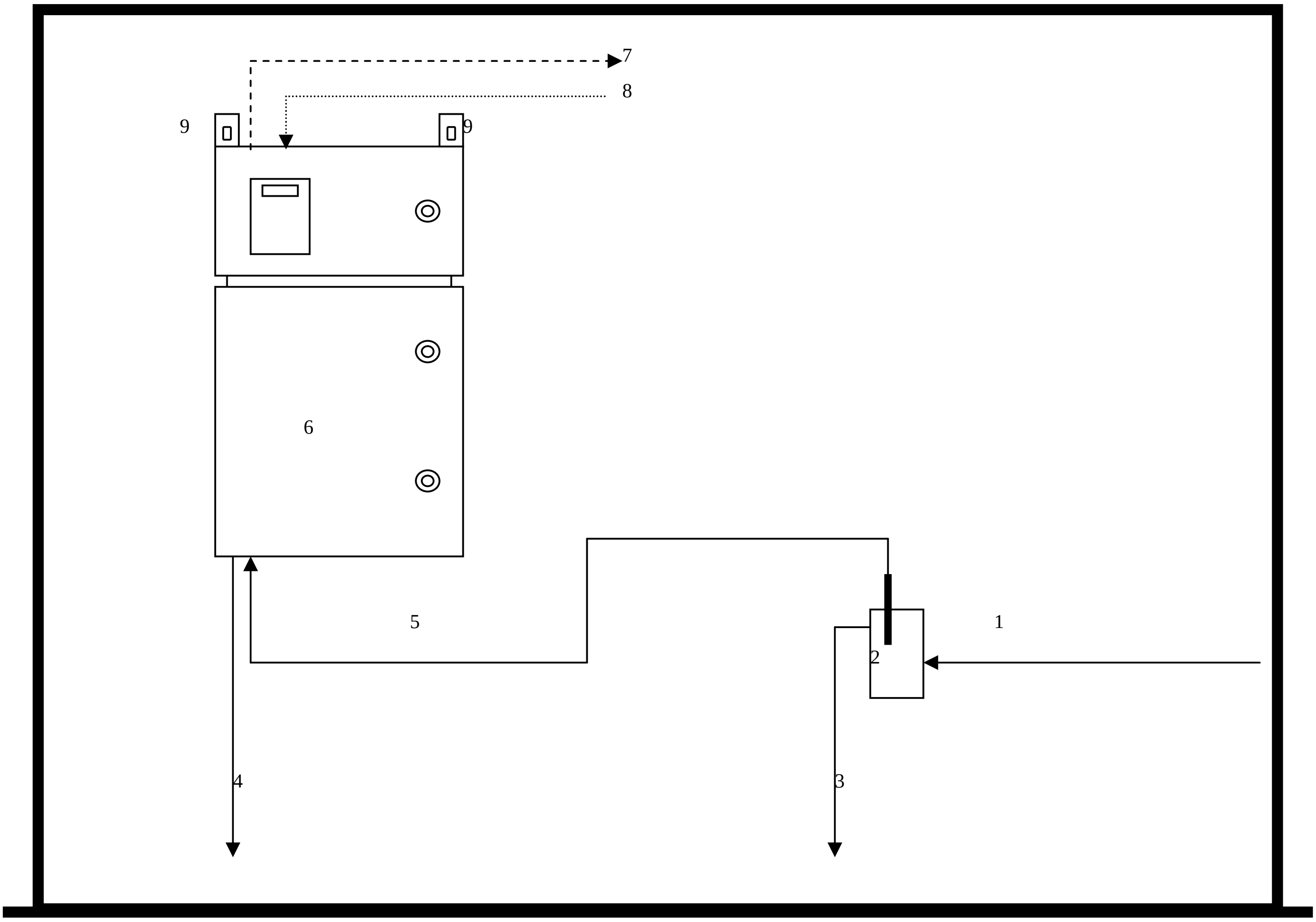


POSIT.	DESCRIPTION
1	POWER IN
2	SIGNALS OUT
3	SAMPLE IN
4	WASTE

SECT	DESCRIPION
A	TOP
B	BOTTOM

3.3 Mounting - schematics

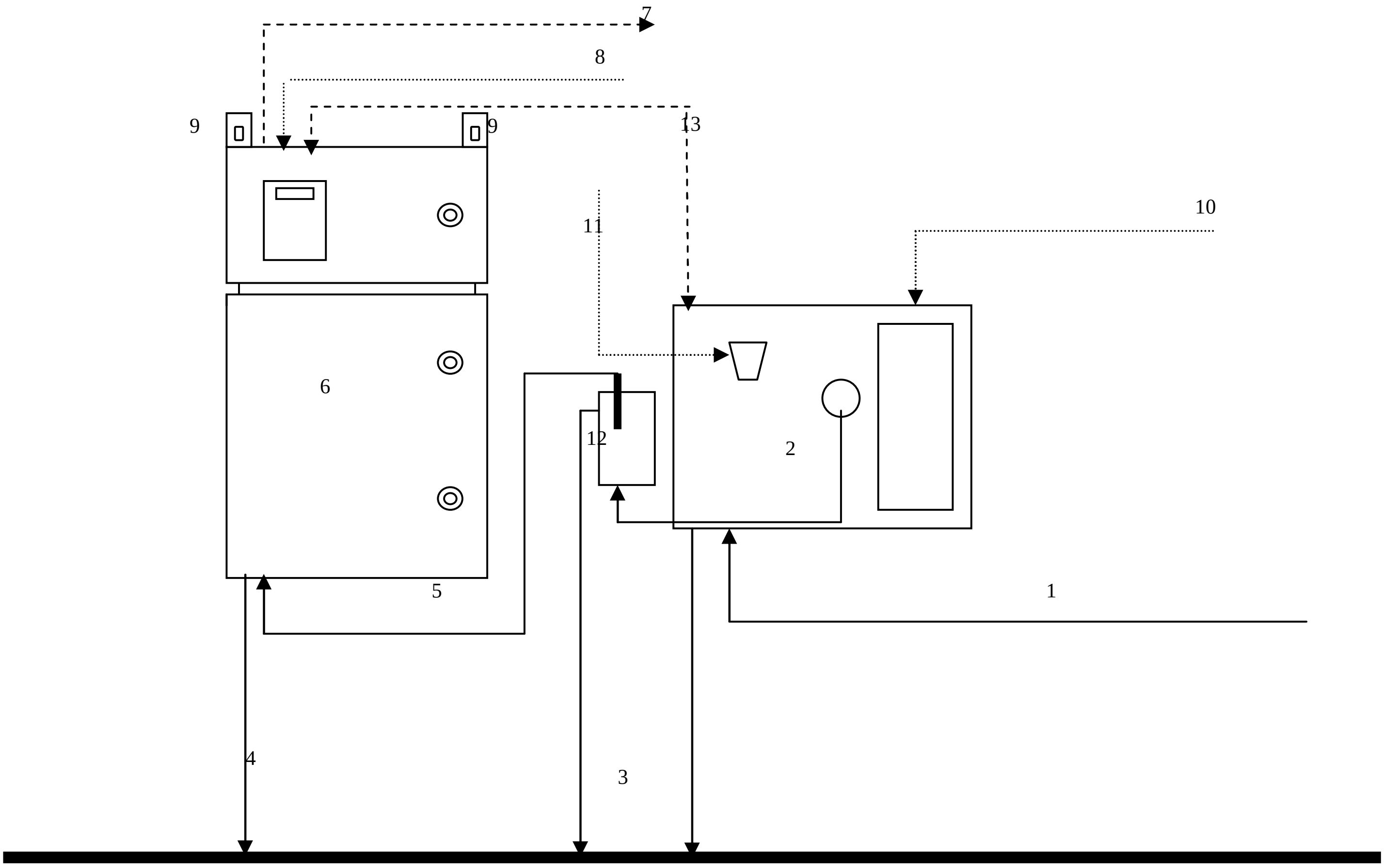
3.3.1 Analyzer mounting w/o filter



Note: Sample delivery Sample overflos is at customer care

Position	Description	Position	Description
1	Sample line	6	Micromac C
2	Sample overflow	7	Alarms, signals output and signals input
3	Overflow waste	8	12 V dc line
4	Analyzer waste silicone 2x 4 mm	9	Wall mounting points
5	Sample inlet to analyzer silicone 2 x 4 mm		

3.3.2 Analyzer mounting with on line self cleaning filter



Position	Description	Position	Description
1	Sample line	7	Alarms, signals output and signals input
2	On line self Cleaning filter	8	12 V dc line
3	Overflow waste	9	Wall mounting points
4	Analyzer waste silicone 2 x 4 mm	10	12 V Dc line
5	Sample inlet to analyzer silicone 2 x 4 mm	11	Compressed air
6	Micromac C	12	Filtered sample pot
		13	Busy contact connection to TB 8 and 9 (see On Line filter manual)

Note: Sample delivery to filter inlet is at customer care

3.4 Environmental requirements

The system should be operated with the ambient temperature ranging from 10 °C to 40 °C.

A temperature near 0 °C freeze the reagents and the calibrant; a temperature over 40 °C will reduce the reagents stability and life, a Peltier cooled reagents compartment can be provided on request.

3.5 Electrical requirements

µMAC C requires the most safe and commonly diffused 12 VDC power supply; a 220 Vac / 12 Vdc power supply can be provided on request.

The minimum absorption in stand-by is 350 mA, the maximum absorption in operation is 2 A, the mean absorption in operation is 0,7 A.

System is protected against current overload, with a 2 A slow blow fuse placed inside the instrument on the board STA-µMAC-03.

3.6 General check

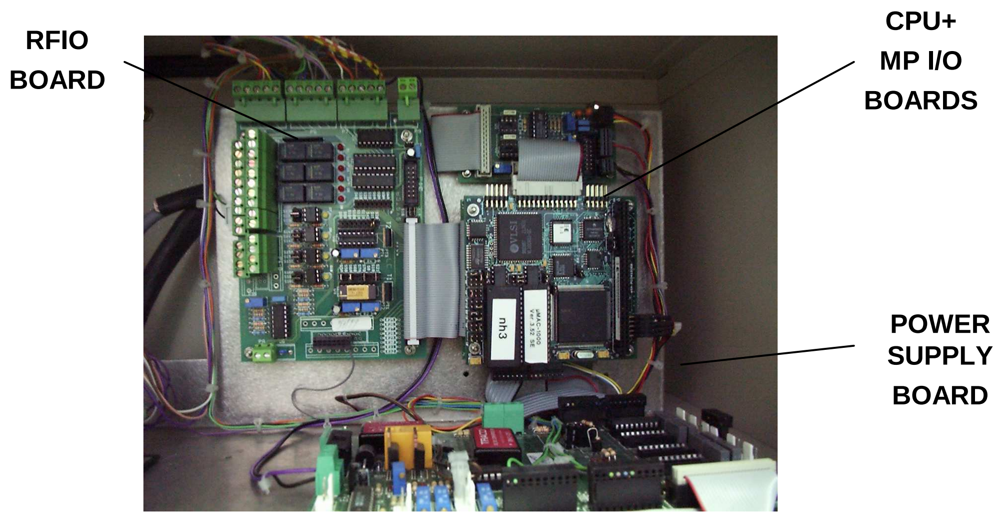
- Before connecting power supply, open both front doors and inspect for any damage; check if everything is well fixed, board connected, connectors inserted.

3.7 Electronic compartment description

μMAC-1000 C MP Electronic compartment

The electronic of μMAC-1000 C MP analyser is composed by the following boards:

- **CPU board**, based on a PC-104 architecture, which manages the analyser with a programmable step-by step program, predefined in factory
- **I/O board**, which manages all the LFR actuation and signal acquisition, using the auxiliary RFIO board
- **RFIO board,** the auxiliary input/output board
- **Power supply board**, which gives to the other boards the necessary DC power supplies.



3.8 Power connection

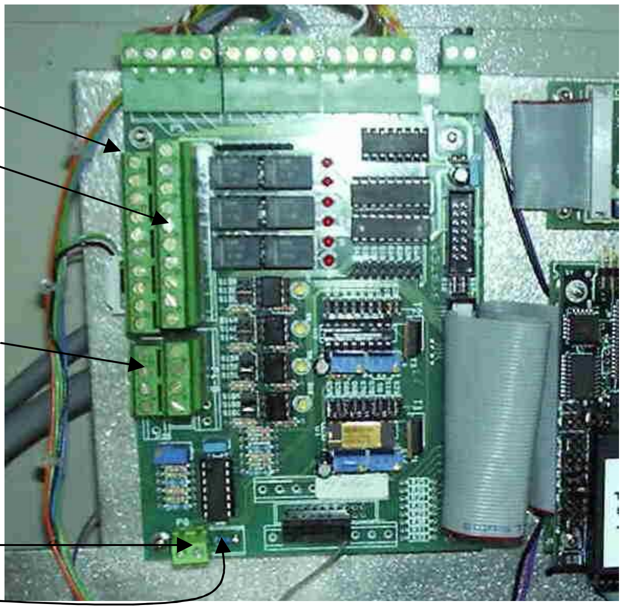
POWER SUPPLY BOARD

Connect the power on the **green connector** of the power board,; take care to connect power cable with right polarity.

For safety reasons it can be useful to measure the current absorption after power on, once connected the AC power; it should be between 350 and 550 mA (if temperature on it can be more than 1 A).

3.9 Signal input/output

RFIO board



3.9.1 **Contact port P3A** (left, method 0)

3.9.2 **Contact port P3B** (right, method 1)

3.9.3 **Input contacts P4A positive** (left row)

Input contacts P4B negative (right row)

3.9.4 **Output analog P8A** (left, method 0)

Output analog P8B (right, method 1)

NOTE: Method 0 or Method 1 can be used by the instrument software also to run a single method analyser for Sample Stream 0 and Sample Stream 1 !

Port	Pin	Internal	Signal type, not act. (actuated)	External use
P3-A	**1**	Relay K1 NC	standby(Busy) Analyser	To recognise if instrument is busy or if it is ready to accept external commands
	2	NO		
	3	COM		
	4	NC	(calibration error) Method 0	To recognise if calibration OD was in or out (=error condition) of set limit
	5	Relay K3 NO		
	6	COM		
	7	NC	(calibration error) Method 1	To recognise if calibration OD was in or out (=error condition) of set limit
	8	Relay K5 NO		
	9	COM		
	10	Relay K7 neg. in	standby(Busy) Method 0	actuation of optional status
	S1	Relay for K7	Standby=not actuated	optional relay for K7

P3-B	**1**	NC	normal(Error) Analyser	To recognise if instrument has produced general Error.
	2	Relay K2 NO		
	3	COM		
	4	NC	(High limit) Method 0	To recognise if measured concentration is in or out (=alarm) of set limit
	5	Relay K4 NO		
	6	COM		
	7	NC	(High limit) Method 1	To recognise if measured concentration is in or out (=alarm) of set limit
	8	Relay K6 NO		
	9	COM		
	10	Relay K8 neg. in	standby(Busy) Method 1	actuation of optional status
	S2	Relay K8	Standby=not actuated	optional relay for K8

P4-A	**1**	Optokoppler +	Start analysis	Remote start of analysis cycle,
P4-B	**1**	5 - 12 VDC -	(jumper S13B)	by switching voltage to OC1
P4-A	**2**	Optokoppler +	Start calibration	Remote start of calibration cycle,
P4-B	**2**	5 - 12 VDC -	(jumper S14B)	by switching voltage to OC2
P4-A	**3**	Optokoppler +	not in use	
P4-B	**3**	5 - 12 VDC -	(jumpers S15A/S15B/S15C)	
P4-A	**4**	Optokoppler +	not in use	
P4-B	**4**	5 - 12 VDC -	(jumpers S16A/S16B/S16C)	
P4-A	**5+6**	GROUND -		Auxiliary supply of external
P4-B	**5+6**	12 VDC +		actuation contacts for P4-A/B

Example: to start the sample from remote, connect **P4-B5** with **P4-A1** (led of OC1 ON)

P8-A	**1**	analog out -	Concentration Method 0	4-20 mA (default)
	2	4-20 mA +	(last measurement)	alternatively 0-20 mA or 0-5 VDC
P8-B	**3**	analog out -	Concentration Method 1	4-20 mA (default)
	4	4-20 mA +	(last measurent)	alternatively 0-20 mA or 0-5 VDC

4. MAINTEINANCE

4.1 Reagent and or calibrant replacement

1. Stop all operations before run any service operation.
2. Check calibrant concentration and measuring unit input.
3. Adit last calibration values (OD Blank and Cal OD) and take note of stored values.
4. Replace reagents and or calibrant solutions, **do not fill old reagents with new solutions**
5. Do not discharge old solutions before completed present procedure
6. Select **Prime** function to fill up reagent lines with new solutions, wait until the analyzer will return in stand by.
7. Select **Calibration** function to run a calibration.
8. Wait until the analyzer will return in stand by.
9. Select **Blank** function to run a reagent blank
10. Wait until the analyzer will return in stand by.
11. Edit the new calibration values (OD Blank and Cal OD) take note of stored values.
12. Start sample analysis

For more information on the reagent replacement procedure and calibration procedure read the software manual on sections:

- Analyzer wash
- Reagents Prime
- Reagent replacement

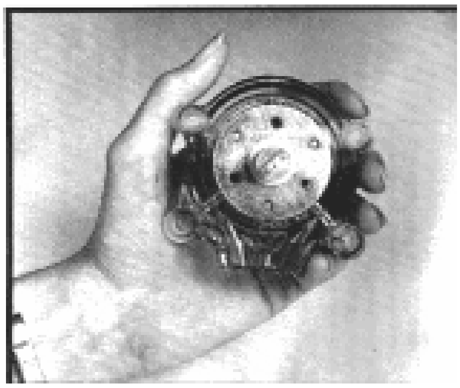
4.2 Pump tube replacement

To be performed every 1500 h of pump operation or every six months.

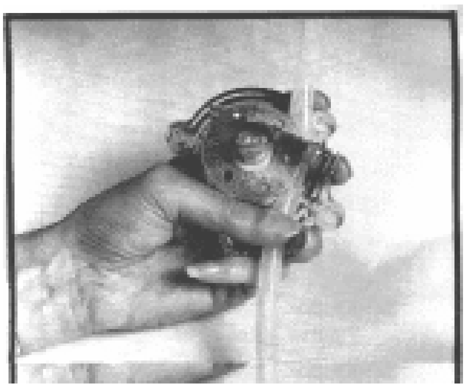
1. Stop any procedure and run a **Wash** to eliminate any trace of reaction products.
2. Switch the system OFF
3. The pump is placed in the lower front side of the instrument. Unscrew manually the four knurled head screws which hold the pump head. The pumping head is now released and can be opened to remove the silicon pump tube.
4. Replace the tube using the same high density i.e. nipple, keeping in mind that the tube that come out from the lower side of the pump must be connected to the S/L valve, and the tube that come out from the upper side must be connected to the bottom nipple of the cylinder 1.

Attention: if the tube is installed in the opposite way, there will be severe malfunctioning of the system!

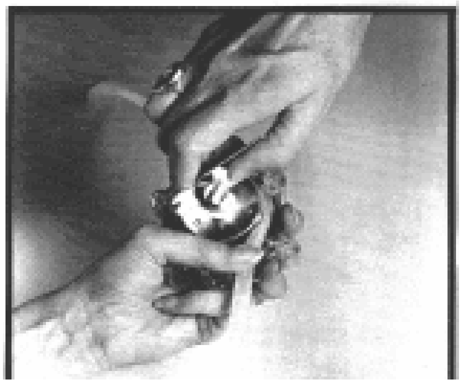
6. Separate pump halves. Hold pump head as shown, with rollers in the 2, 6, and 10 o'clock positions.



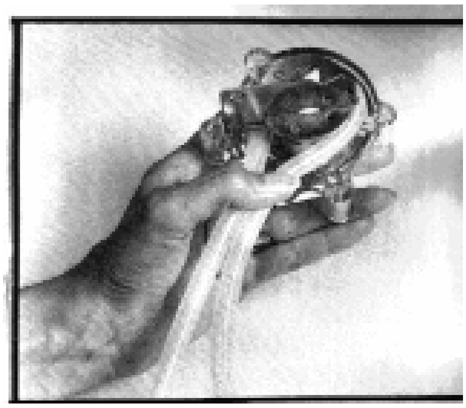
7. Place tubing in the outer port and against the two rollers shown, keeping your thumb on the tubing to hold it in place. Insert tubing key on the back of the rotor shaft and push in as far as possible. Tubing is now positioned deep into the cavity.



8. With the key firmly pressed against the rotor, turn (counterclockwise) and push down while turning, until tubing has surrounded the rotor.



9. Tubing is now in place. Remove key and position other pump half onto the motor shaft and snap shut. Be careful not to pinch tubing between plastic halves.



10. Remount the pump head, taking firmly joined the two halves, moving eventually the roller block until the shaft aligns with the motor drive.
11. Place the four screws, turning right them alternatively, helping with fingers to maintain them in position and checking the right closing of pump head.
12. Switch on the instrument
13. Run a **Wash** and verify that the diluent flows-in the diluent inlet and flows-out from the waste outlet. If not, switch immediately off the instrument and invert the two connections to the pump tube.
    Restart and repeat the check.
14. If there is no aspiration, there should be some connections not properly done, or the pump tube is not mounted correctly.
    Check the connections for any leakage and eventually repeat the pump tube change procedure.

Please note:

1. **The pump tubes have a standard functionality of 1500 h with water. The use of reagents needed for the reactions development may decrease the efficiency.**
2. **The permanence of the system in the STAND BY mode, for some weeks, with no pump movement, may cause the temporary blockage of the pump tube in correspondence of the roll's pressure, blockage which disappear normally after few turns. If the system stop is much more prolonged, in the order of months, it is necessary to remove the tube with the same procedure described for the change, unlock the tube pushing with the fingers. Mount then the tube taking care to the "IN" and "OUT" connections.**
3. **Pump tube deterioration is not sudden, and is considered as one of the factor determining the recalibration frequency. A tube near to its life limit may need a more frequent recalibration.**

4.3 Cleaning procedure

1. Switch off the system.
2. Remove, if any, the cadmium reduction column from the µMAC C unit.
3. Place all the lines, sample, diluent and reagents, in the wash solution.
4. Switch on the system.
5. Run Analysis; check that everything is working properly with special attention to the reagents valves injection; the liquid must flow-out very fast and free from the reagents nipples.
6. In case of injection problem refer to troubleshooting chapter.
7. Place all the lines, samples, diluent and reagents, in distilled water.
8. Run analysis to remove any trace of wash solution from the circuit. Repeat twice.
9. Place the new reagents bottles into the reagent box.
10. Start up the analyzer as described in Software Manual.

Please note:

The cleaning solutions described below are the most commonly used with the µMAC C and must be used when not otherwise specified in the methods description.

CLEANING SOLUTIONS

FOR ACID METHODS:

sodium hypochloride	7%C12	25 ml
sodium hydroxide		10 gr
distilled water	q.s.	250 ml

FOR ALKALINE METHODS:

hydrochloric acid, conc.		10 ml
distilled water	q.s.	250 ml

5. TROUBLESHOOTING

5.1 Hydraulics

5.1.1 Reagents injection

The reagents and the calibrant are injected in the CYLINDER 1 by inhert valves connected to needles passing through the cylinder cover.

Any liquid injected must flow-out from the needle with a fast and thin flow.

In case of partial obstruction of a reagent tube the liquid will flow-out forming small drops.

If the obstruction is total nothing will be injected.

Before the injection the LFR must produce a vacuum, inside the CYLINDER 1, by transferring a liquid volume from CYLINDER 1 to CYLINDER 2.

The minimum level of liquid transfer to obtain a proper vacuum is 1 cm.

5.1.2 TROUBLESHOOTING GUIDE – TABLE 1

TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
INJECTION IN DROPS OR NO INJECTION	LIQUID TRANSFER FOR VACUUM PRODUCTION LESS THAN 1 CM	- CHECK VALVE V6 - CHECK VOLTAGE BELOW 11 VDC - CHANGE PUMP TUBE
	CLOGGING OF THE REAGENT LINE PRESENCE OF PRECIPITATES IN REAGENT CONTAINER	- REMOVE THE REAGENT, PUT THE LINE IN WATER AND START THE WASH PROCEDURE - REPLACE THE REAGENT TUBE
	SILICON PUMP TUBE BLOCKAGE DUE TO A LONG PERIOD (> 10 DAYS) OF INACTIVITY OF THE SYSTEM	- REMOVE THE PUMP TUBE AND CHECK - REPLACE IF NECESSARY
	REAGENT TUBE PINCHED	- CHECK THE REAGENT TUBE FROM REAGENT CONTAINER TO THE REAGENT VALVE, CHECK SLEEVINGS - REPLACE THE REAGENT TUBE
	REAGENT VALVE NOT ACTIVATED ELECTRICALLY	- CHECK THE ELECTRICAL CONNECTIONS
	REAGENT VALVE PROPERLY ACTIVATED BUT NOT OPENING	- CHANGE THE VALVE, - CHECK THE STA-RFIO BOARD AND REPLACE IF NECESSARY

5.1.2 TROUBLESHOOTING GUIDE - TABLE 2

TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
VALUES AFTER DILUITION NEAR ZERO	SILICON TUBE BETWEEN V11 NC BLOCKED	- CHANGE THE TUBE
	DILUTION VALVES NOT SWITCHING	- CHECK THE VALVE WITH AN EXTERNAL 12 VDC POWER SOURCE. - CHECK THE STA-RFIO BOARD AND REPLACE IF NECESSARY
LEAKAGE IN SOME LFR CONNECTION DUE TO OVER PRESSURE	VOLTAGE SUPPLY OVER 13 VDC	-CHECK THE EXTERNAL POWER SUPPLY
	WASTE TUBE PINCHED	- CHECK AND FREE THE TUBE
	VALVE V5 OR V7 NOT ACTIVATED	- CHECK THE CONNECTIONS AND THE VALVE WITH AN EXTERNAL 12 VDC POWER SOURCE. - CHECK THE STA-RFIO BOARD AND REPLACE IF NECESSARY
	VALVE V5 OR V7 CLOGGED	- TRY TO CLEAN WITH A SYRINGE CONTAINING A CLEANING SOLUTION - REPLACE VALVES

5.1.2 TROUBLESHOOTING GUIDE - TABLE 2 (cont'd)

TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
NO VALUES ON SAMPLE BUT CALIBRATION GIVES GOOD RESULT	VALVE VC5 DEFECTIVE	- CHECK THAT DURING I SAMPLING THE LIQUID IS ASPIRATED ONLY FROM THE SAMPLE LINE - CHECK THAT DURING SAMPLING NO LIQUID IS ASPIRATED FROM THE WASH AND CALIBRANT LINES - CHECK THE CONNECTIONS AND THE VALVE WITH AN EXTERNAL 12 VDC POWER SOURCE. - CHECK THE STA-RFIO BOARD AND REPLACE IF NECESSARY
CALIBRATION VALUE IS NOT REPRODUCIBLE	VALVE V8 OR VC5 DEFECTIVE	- CHECK THAT DURING CALIBRANT SAMPLING LIQUID IS ASPIRATED ONLY FORM THE CALIBRATION LINE - CHECK THAT DURING SAMPLING NO LIQUID IS ASPIRATED FROM THE WASH AND SAMPLE LINES - CHECK THE CONNECTIONS AND THE VALVE WITH AN EXTERNAL 12 VDC POWER SOURCE. - CHECK THE STA-RFIO BOARD AND REPLACE IF NECESSARY
BAD VALUES ON SAMPLE AND CALIBRATION	ONE OR MORE REAGENTS NOT INJECTED PROPERLY	- RUN A REAGENT PRIME AND CHECK THAT ALL VALVES INJECT PROOPERLY - RUN THE CLEANING PROCEDURE

6. SPARES LIST

P/N	Description	Sales unit	Module	Method	Type	Suggested stock
116-0536-10	TUBING, T. 0.045	Mt.	MANIFOLD	All	Spare	1
116-0536-15	TUBING, T. 0.081	Mt.	MANIFOLD	All	Spare	1
116-0537-10	TUBING, S. 0.045	Mt.	MANIFOLD	All	Spare	1
116-CFLEX-01	C FLEX TUBING SMALL 1st BUSHING for 562-3019-X1 (6424-50)	Mt.	MANIFOLD	All	Spare	1
116-CFLEX-02	C FLEX TUBING BIGF 2nd BUSHING 562-3019-x1->178-6275-03 (6424-62)	Mt.	MANIFOLD	All	Spare	1
116-R497-110	TRANSMISSION TUBE, SILICONE 0,045	Mt.	MANIFOLD	All	Spare	1
116-R497-150	TRANSMISSION TUBE, SILICONE 0,065	Mt.	MANIFOLD	All	Spare	1
178-6275-03	REAGENT STRAW TEFLON 1,6X32	PK/10	MANIFOLD	All	Spare	1
562-3019-X1	TUBING TEFLON 1 x 1,5 mm (LFA reag. Tube)	Mt.	MANIFOLD	All	Spare	5
562-3019-X2	TUBING TEFLON 1.6 x 3.2 mm (NPA+ Acidic manifold)	Mt.	MANIFOLD	COD, TP, Total Iron, strong acidic methods	Spare	10
562-3050-X2	TUBING SILICONE 1.8X4.0MM	Mt.	MANIFOLD	All	Spare	10
CPU-DISK-00	DISK ON CHIP FOR CPU-PC104-00	EA/Unità	HARDWARE		On demand	
CPU-PC104-00	CPU BOARD WITHOUT DISK ON CHIP	EA/Unità	HARDWARE		On demand	
MM1-0459-00	T CONNECTOR - KARTELL	EA/Unità	MANIFOLD	All	Spare	5
MM1-1250-00	L CONNECTOR - KARTELL	EA/Unità	MANIFOLD	All	Spare	5
PRS-SNSR-01	SAMPLE PRESENCE DETECTOR	EA/Unità	HARDWARE	All	On demand	
QRZ-COIL-02	QUARTZ COILS UV DIGESTOR 2 mm ID X 4 mm (LFA)	EA/Unità	YDRAULICS	TP, TN; All methods using UV digestor	Spare	1

P/N	Description	Sales unit	Module	Method	Type	Suggested stock
STA-BATH-01	HEATING BATH - LFA ANALYZER (TEFLON)	EA/Unità	MANIFOLD	Silica, NO3 Hydrazine, Tot P, Tot N, Tot Fe	On demand	
STA-BATH-15	HEATING BATH FLOW CELL & 15 MM FLOW CELL ASSY	EA/Unità	COLORIM.	Ammonia Standard range	On demand	
STA-BATH-50	HEATING BATH FLOW CELL & 50 MM FLOW CELL ASSY	EA/Unità	COLORIM.	Ammonia Seawater range	On demand	
STA-CDCL-01	Cd COLUMN LFA ANALYZERS, FILLED	EA/Unità	MANIFOLD	NO3 Cd reduction	Consumable	2
STA-DISP-02	LFA ANALYZER DISPLAY, PARALLEL/GRAPHICS	EA/Unità	ARDWARE		On demand	
STA-DISPASSY-01	LFA ANALYZ. DISPLAY ASSY-PAR/GRAPH (KEYP+BD+DISP)	EA/Unità	HARDWARE		On demand	
STA-FER-02	FERULA SGM TEFLON 1,6x3,2 (Manifold valve)	EA/Unità	MANIFOLD	COD, TP, Strong acidic methods	Spare	5
STA-FER-03	FERULA TEFLON 1,0x1,5 (Reagent Valve)	EA/Unità	MANIFOLD	COD, TP, Strong acidic methods	Spare	5
STA-KEYP-02	LFA ANALYZER KEYPAD, NEW WITH LOGO	EA/Unità	ARDWARE		On demand	
STA-M100-00	REACTION CYLINDER BLOCK	EA/Unità	HYDRAULIC	All	Spare	1
STA-M100-01	FLOC CELL HOLDER (UNIVERSAL)	EA/Unità	COLORIM.		On demand	
STA-M101-01	FLOW CELL 50 mm	EA/Unità	COLORIM.	Seawater applications, High sensitivity methods; TP, TN	Spare	1
STA-M101-02	FLOW CELL 15 mm	EA/Unità	COLORIM.	All Standars applications	Spare	1
STA-M101-09	FLOW CELL 5 mm	EA/Unità	COLORIM.	High level applications	Spare	1
STA-M103-01	PUMP MOTOR	EA/Unità	HYDRAULIC	All	On demand	
STA-M103-02	PUMP HEAD	EA/Unità	HYDRAULIC	All	Spare	1

P/N	Description	Sales unit	Module	Method	Type	Suggested stock
STA-M103-03	PUMP TUBE, MICROMAC	EA/Unità	MANIFOLD	All	Consumable	3
STA-M103-05	TEFLON PUMP COD REAGENT INJECTION	EA/Unità	MANIFOLD	COD	On demand	
STA-M105-05	2 WAY VALVE NEW (N.0.)	EA/Unità	MANIFOLD	All	Spare	1
STA-M105-06	3 WAY VALVE NEW	EA/Unità	MANIFOLD	All	Spare	2
STA-M105-07	2 WAY VALVE NEW (N.C.)	EA/Unità	MANIFOLD	All	Spare	1
STA-M105-12	2 WAY VALVE (SMC)	EA/Unità	MANIFOLD	Reagent valve all analyzers		1
STA-M105-14	3 WAY VALVE TEFLON (MANIFOLD)	EA/Unità	MANIFOLD	COD, TP, Strong Acidic methods	Spare	2
STA-M105-15	2 WAY VALVE TEFLON (REAGENT INJECTION)	EA/Unità	MANIFOLD	COD, TP, Strong Acidic methods	Spare	2
STA-M200-01	SAMPLE POT MICROMAC C	EA/Unità	HYDRAULIC	All	On demand	
STA-MCOL-00	COLORIMETER PREAMPLIFIER BOARD	EA/Unità	HARDWARE		On demand	
STA-MPIO-01	I/O BOARD PC 104	EA/Unità	HARDWARE		On demand	
STA-NUT-01	NUT TEFLON 1,6x3,2 - BLACK (Manifold valve)	EA/Unità	MANIFOLD	COD, TP, Strong acidic methods	Spare	5
STA-NUT-03	NUT TEFLON 1,0x1,5 (Reagent valve)	EA/Unità	MANIFOLD	NPA, COD, TP, Strong acidic methods	Spare	5
STA-ORING-01	O-RING RILSAN FOR STA-M105-02	EA/Unità	HYDRAULIC		Spare	5
STA-RFIO-04	DRIVER BOARD NEW (Micromac 1000)	EA/Unità	HARDWARE			
STA-RFIO-04C	DRIVER BOARD NEW (Micromac C)	EA/Unità	HARDWARE		On demand	
STA-TSNS-01	H.B OR F/C TEMPERATURE SENSOR LM35 WITH CABLE	EA/Unità	COLORIM.	NH3, SiO2, NO3 Hydrazine, TP, TN Tot Fe etc.	On demand	

P/N	Description	Sales unit	Module	Method	Type	Suggested stock
STA-uMAC-03E	POWER SUPPLY DRIVER - NEW (graphic/parallel display)	EA/Unità	HARDWARE		On demand	
STA-uMAC-06	KEYBOARD INTERFACE BOARD; PARALL/GRAPH DISPLAY	EA/Unità	HARDWARE		On demand	
STA-UVLM-01	UV LAMP LFA/FLOWSYS	EA/Unità	MANIFOLD	TP, TN; All methods using UV digestor	Consumable	2
STA-UVPW-01	UV DIGESTOR STARTER	EA/Unità	HARDWARE	TP, TN; All methods using UV digestor	Spare	1
STA-WELL-01	HEATING BATH RESISTOR	EA/Unità	MANIFOLD		On demand	
STA-XTR110	ELECTRONIC COMPONENT4- 20 mA	EA/Unità	HARDWARE		On demand	